# Lecture Notes on the Ratio $|GL_m(\mathbb{F}_q)|/|\mathrm{Mat}_m(\mathbb{F}_q)|$

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## 1 Introduction

The goal of these notes is to understand the ratio

$$\frac{|GL_m(\mathbb{F}_q)|}{|\,\mathrm{Mat}_m(\mathbb{F}_q)|}.$$

Here:

- $\mathbb{F}_q$ is the finite field with $q$ elements,
- $\mathrm{Mat}_m(\mathbb{F}_q)$ denotes the set of all $m \times m$ matrices over $\mathbb{F}_q$,
- $GL_m(\mathbb{F}_q)$ denotes the set of all invertible $m \times m$ matrices over $\mathbb{F}_q$.

This ratio has a simple probabilistic meaning:

If one chooses a matrix uniformly at random from $\mathrm{Mat}_m(\mathbb{F}_q)$, then

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|}$$

is exactly the probability that the matrix is invertible.

So our aim is to compute this ratio explicitly and understand what it means.

## 2 The ambient set of all matrices

**Definition 2.1.** For $m \in \mathbb{Z}_{>0}$, let

$$\mathrm{Mat}_m(\mathbb{F}_q) = \mathbb{F}_q^{m\times m}$$

denote the set of all $m \times m$ matrices over $\mathbb{F}_q$.

**Proposition 2.2.** *The number of $m \times m$ matrices over $\mathbb{F}_q$ is*

$$|\mathrm{Mat}_m(\mathbb{F}_q)| = q^{m^2}.$$

*Proof.* An $m \times m$ matrix has $m^2$ entries, and each entry can be chosen independently from $\mathbb{F}_q$, which has $q$ elements. Hence the total number of such matrices is

$$\underbrace{q \cdot q \cdot \cdots \cdot q}_{m^2 \text{ factors}} = q^{m^2}.$$

□

**Remark 2.3.** Thus $\mathrm{Mat}_m(\mathbb{F}_q)$ is the full space of all linear endomorphisms $\mathbb{F}_q^m \to \mathbb{F}_q^m$, while $GL_m(\mathbb{F}_q)$ is the subset of those endomorphisms that are invertible.

## 3 The general linear group

**Definition 3.1.** The *general linear group* over $\mathbb{F}_q$ is

$$GL_m(\mathbb{F}_q) = \{A \in \mathrm{Mat}_m(\mathbb{F}_q) : A \text{ is invertible}\}.$$

**Remark 3.2.** A matrix $A \in \mathrm{Mat}_m(\mathbb{F}_q)$ is invertible if and only if the associated linear map

$$T_A : \mathbb{F}_q^m \to \mathbb{F}_q^m, \qquad x \mapsto Ax$$

is bijective.

## 4 Why invertibility is equivalent to linear independence of columns

Let

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_m \\ | & | & & | \end{pmatrix} \in \mathrm{Mat}_m(\mathbb{F}_q),$$

where $v_1, \ldots, v_m \in \mathbb{F}_q^m$ are the columns of $A$.

**Proposition 4.1.** *The matrix $A$ is invertible if and only if its columns $v_1, \ldots, v_m$ are linearly independent.*

*Proof.* Consider the linear map

$$T_A : \mathbb{F}_q^m \to \mathbb{F}_q^m, \qquad x \mapsto Ax.$$

Since $A$ is a square matrix, $A$ is invertible if and only if $T_A$ is bijective.

Now let

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} \in \mathbb{F}_q^m.$$

Then

$$Ax = x_1 v_1 + \cdots + x_m v_m.$$

Therefore

$$Ax = 0 \quad \Longleftrightarrow \quad x_1 v_1 + \cdots + x_m v_m = 0.$$

So the columns $v_1, \ldots, v_m$ are linearly independent if and only if the only solution to $Ax = 0$ is $x = 0$, that is,

$$\mathrm{Ker}(A) = \{0\}.$$

Since $A : \mathbb{F}_q^m \to \mathbb{F}_q^m$ is a linear map between vector spaces of the same dimension, $\mathrm{Ker}(A) = \{0\}$ is equivalent to injectivity, and injectivity is equivalent to bijectivity. Hence $A$ is invertible if and only if its columns are linearly independent. □

# 5 Counting invertible matrices

**Theorem 5.1.** *Let $m \in \mathbb{Z}_{>0}$. Then*

$$|GL_m(\mathbb{F}_q)| = (q^m - 1)(q^m - q)(q^m - q^2) \cdots (q^m - q^{m-1}).$$

*Equivalently,*

$$|GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i).$$

*Proof.* We count invertible matrices column by column.

An $m \times m$ matrix is invertible if and only if its columns are linearly independent.

**First column.** The first column may be any nonzero vector in $\mathbb{F}_q^m$. Since $\mathbb{F}_q^m$ has $q^m$ vectors total and exactly one of them is zero, there are

$$q^m - 1$$

choices.

**Second column.** Once the first column has been chosen, the second column must not lie in its span. Since the span of a nonzero vector is a 1-dimensional subspace of $\mathbb{F}_q^m$, it has exactly $q$ elements. Therefore there are

$$q^m - q$$

choices for the second column.

**Third column.** Once the first two columns are linearly independent, their span is a 2-dimensional subspace of $\mathbb{F}_q^m$, hence has exactly $q^2$ elements. Therefore there are

$$q^m - q^2$$

choices for the third column.

Continuing in this way, once $i$ linearly independent columns have been chosen, their span is an $i$-dimensional subspace of $\mathbb{F}_q^m$, so it has exactly $q^i$ elements. Therefore the next column can be chosen in

$$q^m - q^i$$

ways.

Multiplying all these choices gives

$$|GL_m(\mathbb{F}_q)| = (q^m - 1)(q^m - q)(q^m - q^2) \cdots (q^m - q^{m-1}).$$

□

## 6 The ratio $|GL_m(\mathbb{F}_q)|/|\mathrm{Mat}_m(\mathbb{F}_q)|$

Now we divide the number of invertible matrices by the number of all matrices.

**Theorem 6.1.** *For every $m \geq 1$,*

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|} = \frac{(q^m - 1)(q^m - q)(q^m - q^2) \cdots (q^m - q^{m-1})}{q^{m^2}}.$$

*Equivalently,*

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m} (1 - q^{-j}).$$

*Proof.* From the previous sections,

$$|GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i) \qquad \text{and} \qquad |\mathrm{Mat}_m(\mathbb{F}_q)| = q^{m^2}.$$

Hence

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|} = \frac{\prod_{i=0}^{m-1}(q^m - q^i)}{q^{m^2}}.$$

Now factor $q^m$ out of each term:

$$q^m - q^i = q^m(1 - q^{i-m}).$$

Therefore

$$\prod_{i=0}^{m-1} (q^m - q^i) = \prod_{i=0}^{m-1} q^m(1 - q^{i-m}) = q^{m^2} \prod_{i=0}^{m-1} (1 - q^{i-m}).$$

Dividing by $q^{m^2}$ gives

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|} = \prod_{i=0}^{m-1} (1 - q^{i-m}).$$

If we reindex by setting $j = m - i$, then $j$ runs from 1 to $m$, and we obtain

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m} (1 - q^{-j}).$$

□

## 7 Probabilistic interpretation

**Proposition 7.1.** *If a matrix is chosen uniformly at random from* $\mathrm{Mat}_m(\mathbb{F}_q)$*, then*

$$\mathbb{P}(\textit{the matrix is invertible}) = \frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1 - q^{-j}).$$

*Proof.* There are $|\mathrm{Mat}_m(\mathbb{F}_q)|$ matrices total, and exactly $|GL_m(\mathbb{F}_q)|$ of them are invertible. Since each matrix is chosen with equal probability, the probability of choosing an invertible matrix is

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|}.$$

The explicit product formula was established above. □

**Remark 7.2.** Thus the ratio

$$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|}$$

is not just an abstract quotient of cardinalities. It is the density of invertible matrices inside the space of all square matrices.

## 8 A second derivation by conditional probability

There is another way to obtain the same formula, directly as a probability computation.

**Theorem 8.1.** *Let $A$ be a uniformly random matrix in* $\mathrm{Mat}_m(\mathbb{F}_q)$*. Then*

$$\mathbb{P}(A \in GL_m(\mathbb{F}_q)) = \prod_{j=1}^{m}(1 - q^{-j}).$$

*Proof.* Write the columns of $A$ as $v_1, \ldots, v_m \in \mathbb{F}_q^m$. The matrix $A$ is invertible if and only if these columns are linearly independent.

**Step 1.** The probability that $v_1 \neq 0$ is

$$\frac{q^m - 1}{q^m} = 1 - q^{-m}.$$

**Step 2.** Given that $v_1 \neq 0$, the vector $v_2$ must avoid $\mathrm{span}(v_1)$, which has $q$ elements. Hence

$$\mathbb{P}(v_2 \notin \mathrm{span}(v_1) \mid v_1 \neq 0) = \frac{q^m - q}{q^m} = 1 - q^{1-m}.$$

**Step 3.** Given that $v_1, v_2$ are linearly independent, the vector $v_3$ must avoid $\mathrm{span}(v_1, v_2)$, which has $q^2$ elements. Hence

$$\mathbb{P}(v_3 \notin \mathrm{span}(v_1, v_2) \mid v_1, v_2 \text{ independent}) = \frac{q^m - q^2}{q^m} = 1 - q^{2-m}.$$

Continuing this way, the probability that the $k$-th column avoids the span of the first $k-1$ independent columns is

$$1 - q^{k-1-m}.$$

Multiplying these conditional probabilities, we obtain

$$\mathbb{P}(A \in GL_m(\mathbb{F}_q)) = \prod_{k=1}^{m}(1 - q^{k-1-m}).$$

Reindexing gives

$$\mathbb{P}(A \in GL_m(\mathbb{F}_q)) = \prod_{j=1}^{m} (1 - q^{-j}).$$

$\square$

# 9 Examples

## 9.1 Case $m = 1$

We have

$$|GL_1(\mathbb{F}_q)| = q - 1 \qquad \text{and} \qquad |\operatorname{Mat}_1(\mathbb{F}_q)| = q.$$

So

$$\frac{|GL_1(\mathbb{F}_q)|}{|\operatorname{Mat}_1(\mathbb{F}_q)|} = \frac{q-1}{q} = 1 - q^{-1}.$$

## 9.2 Case $m = 2$

We have

$$|GL_2(\mathbb{F}_q)| = (q^2 - 1)(q^2 - q) \qquad \text{and} \qquad |\operatorname{Mat}_2(\mathbb{F}_q)| = q^4.$$

Therefore

$$\frac{|GL_2(\mathbb{F}_q)|}{|\operatorname{Mat}_2(\mathbb{F}_q)|} = \frac{(q^2-1)(q^2-q)}{q^4} = (1 - q^{-1})(1 - q^{-2}).$$

## 9.3 Case $m = 3$

We have

$$|GL_3(\mathbb{F}_q)| = (q^3 - 1)(q^3 - q)(q^3 - q^2) \qquad \text{and} \qquad |\operatorname{Mat}_3(\mathbb{F}_q)| = q^9.$$

Hence

$$\frac{|GL_3(\mathbb{F}_q)|}{|\operatorname{Mat}_3(\mathbb{F}_q)|} = (1 - q^{-1})(1 - q^{-2})(1 - q^{-3}).$$

**Example 9.1.** Over $\mathbb{F}_2$,

$$\frac{|GL_2(\mathbb{F}_2)|}{|\operatorname{Mat}_2(\mathbb{F}_2)|} = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right) = \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}.$$

Since $|\operatorname{Mat}_2(\mathbb{F}_2)| = 2^4 = 16$, this means that exactly

$$16 \cdot \frac{3}{8} = 6$$

matrices are invertible, which agrees with

$$|GL_2(\mathbb{F}_2)| = (4 - 1)(4 - 2) = 6.$$

**Example 9.2.** Over $\mathbb{F}_2$,

$$\frac{|GL_3(\mathbb{F}_2)|}{|\operatorname{Mat}_3(\mathbb{F}_2)|} = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{8}\right) = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{7}{8} = \frac{21}{64}.$$

Since $|\operatorname{Mat}_3(\mathbb{F}_2)| = 2^9 = 512$, it follows that

$$|GL_3(\mathbb{F}_2)| = 512 \cdot \frac{21}{64} = 168.$$

## 10 Asymptotic viewpoint

For fixed $q$, the ratio

$$\frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1-q^{-j})$$

has a limit as $m \to \infty$:

$$\lim_{m\to\infty} \frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{\infty}(1-q^{-j}).$$

**Remark 10.1.** This means that even for very large matrices over a fixed finite field, the probability of being invertible does not go to 0. Instead, it approaches a positive constant.

**Example 10.2.** For $q = 2$, the limiting probability is

$$\prod_{j=1}^{\infty}(1-2^{-j}) \approx 0.288\ldots$$

So a large random binary matrix still has a substantial chance of being invertible.

## 11 Conceptual summary

We can summarize the story in one chain of ideas:

invertible matrix $\iff$ columns are linearly independent $\iff$ choose columns avoiding previous spans.

This leads to the counting formula

$$|GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m - q^i).$$

Since

$$|\operatorname{Mat}_m(\mathbb{F}_q)| = q^{m^2},$$

the ratio becomes

$$\frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1-q^{-j}).$$

Thus the ratio has three equivalent meanings:

(1) it is a quotient of cardinalities,

(2) it is the density of invertible matrices inside all square matrices,

(3) it is the probability that a random $m \times m$ matrix over $\mathbb{F}_q$ is invertible.

## Exercises

1. Compute

$$\frac{|GL_1(\mathbb{F}_q)|}{|\operatorname{Mat}_1(\mathbb{F}_q)|}$$

directly.

2. Verify that
$$\frac{|GL_2(\mathbb{F}_q)|}{|\operatorname{Mat}_2(\mathbb{F}_q)|} = (1-q^{-1})(1-q^{-2}).$$

3. Show that for $q = 3$,
$$\frac{|GL_2(\mathbb{F}_3)|}{|\operatorname{Mat}_2(\mathbb{F}_3)|} = \left(1-\frac{1}{3}\right)\left(1-\frac{1}{9}\right).$$

4. Prove directly that
$$|GL_4(\mathbb{F}_q)| = (q^4-1)(q^4-q)(q^4-q^2)(q^4-q^3).$$

5. Explain why
$$\frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|}$$
is the probability that a random matrix in $\operatorname{Mat}_m(\mathbb{F}_q)$ is invertible.

## 12 Table and graph of the ratio $|GL_m(\mathbb{F}_q)|/|\mathrm{Mat}_m(\mathbb{F}_q)|$

Recall that

$$|\,\mathrm{Mat}_m(\mathbb{F}_q)| = q^{m^2}, \qquad |GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m - q^i),$$

and hence

$$\frac{|GL_m(\mathbb{F}_q)|}{|\,\mathrm{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1 - q^{-j}).$$

### 12.1 A table for small values of $m$

| $m$ | $\lvert\,\mathrm{Mat}_m(\mathbb{F}_q)\rvert$ | $\lvert GL_m(\mathbb{F}_q)\rvert$ | $\dfrac{\lvert GL_m(\mathbb{F}_q)\rvert}{\lvert\,\mathrm{Mat}_m(\mathbb{F}_q)\rvert}$ |
|---|---|---|---|
| 1 | $q$ | $q-1$ | $1-q^{-1}$ |
| 2 | $q^4$ | $(q^2-1)(q^2-q)$ | $(1-q^{-1})(1-q^{-2})$ |
| 3 | $q^9$ | $(q^3-1)(q^3-q)(q^3-q^2)$ | $(1-q^{-1})(1-q^{-2})(1-q^{-3})$ |
| 4 | $q^{16}$ | $(q^4-1)(q^4-q)(q^4-q^2)(q^4-q^3)$ | $(1-q^{-1})(1-q^{-2})(1-q^{-3})(1-q^{-4})$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| $m$ | $q^{m^2}$ | $\prod_{i=0}^{m-1}(q^m-q^i)$ | $\prod_{j=1}^{m}(1-q^{-j})$ |

### 12.2 A graph of the ratio as a function of $m$

Fix $q$. For illustration, the following graph shows

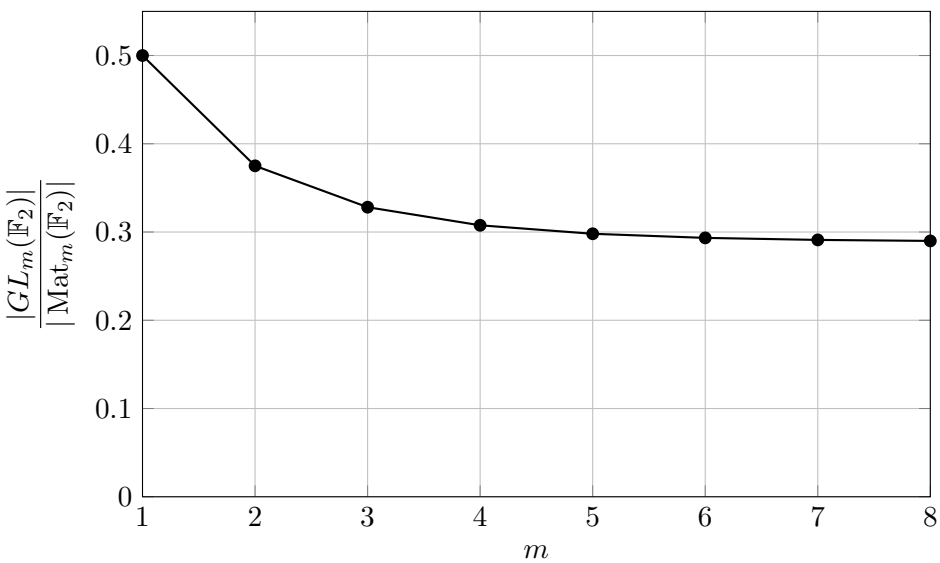
$$y = \frac{|GL_m(\mathbb{F}_q)|}{|\,\mathrm{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1 - q^{-j})$$

as a function of $m$.

For example, if we take $q = 2$, then the first few values are

$$\frac{|GL_1(\mathbb{F}_2)|}{|\,\mathrm{Mat}_1(\mathbb{F}_2)|} = \frac{1}{2}, \qquad \frac{|GL_2(\mathbb{F}_2)|}{|\,\mathrm{Mat}_2(\mathbb{F}_2)|} = \frac{3}{8}, \qquad \frac{|GL_3(\mathbb{F}_2)|}{|\,\mathrm{Mat}_3(\mathbb{F}_2)|} = \frac{21}{64},$$

and so on.



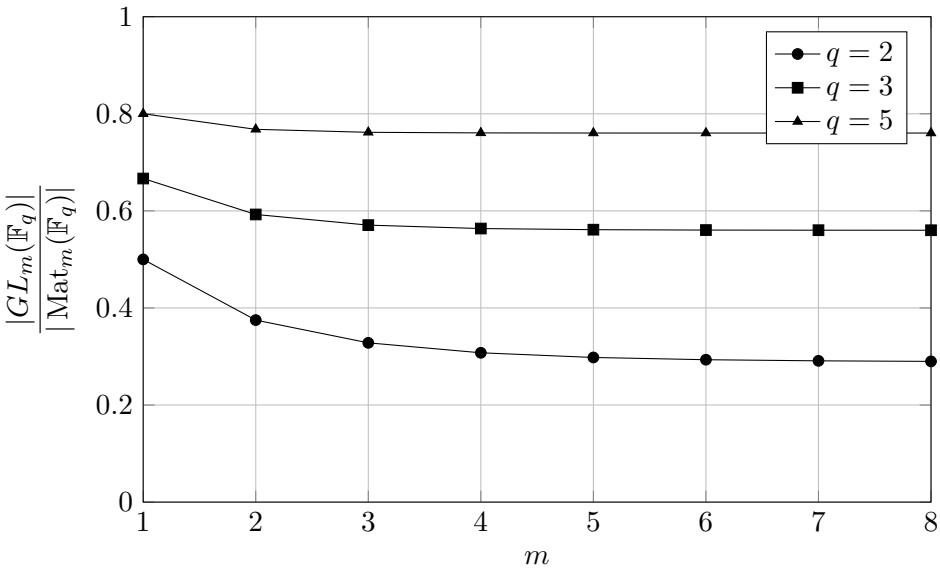
**Remark 12.1.** For fixed $q$, the ratio

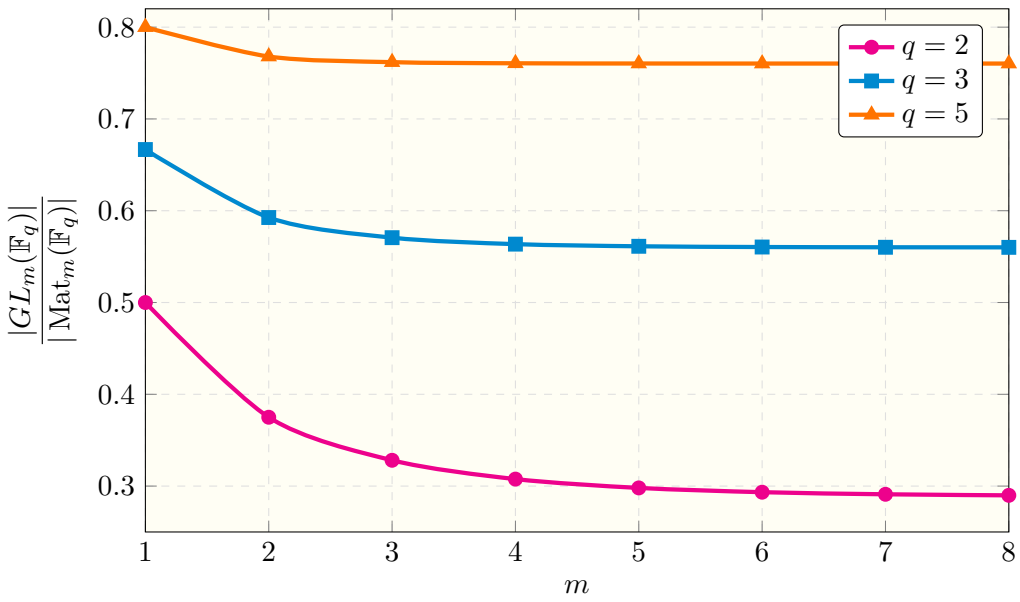
$$\frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1 - q^{-j})$$

decreases as $m$ increases and converges to the infinite product

$$\prod_{j=1}^{\infty}(1 - q^{-j}).$$

For $q = 2$, this limit is approximately 0.288.




0.8
0.7
0.6
0.5
0.4
0.3
$\frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|}$
1
2
3
4
5
6
7
8
$m$
$q = 2$
$q = 3$
$q = 5$

**Proposition 12.2.** *Let $q \geq 2$ be fixed, and define*

$$a_m := \frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1 - q^{-j}).$$

*Then the sequence $(a_m)_{m\geq 1}$ converges as $m \to \infty$. Moreover, its limit is positive.*

*Proof.* For each $j \geq 1$, we have

$$0 < 1 - q^{-j} < 1.$$

Hence

$$0 < a_m \leq 1$$

for all $m \geq 1$.

Also,

$$a_{m+1} = a_m(1 - q^{-(m+1)}),$$

and since $0 < 1 - q^{-(m+1)} < 1$, it follows that

$$a_{m+1} < a_m.$$

Thus $(a_m)$ is a decreasing sequence of positive real numbers. Therefore it converges to some limit

$$L \geq 0.$$

It remains to show that $L > 0$. Since $q \geq 2$, we have $0 < q^{-j} \leq 1/2$ for all $j \geq 1$. Using the estimate

$$\log(1 - x) \geq -2x \qquad \text{for } 0 \leq x \leq \tfrac{1}{2},$$

we obtain

$$\log(1 - q^{-j}) \geq -2q^{-j}.$$

Therefore

$$\log a_m = \sum_{j=1}^{m} \log(1 - q^{-j}) \geq -2\sum_{j=1}^{m} q^{-j}.$$

Since the geometric series

$$\sum_{j=1}^{\infty} q^{-j}$$

converges, the right-hand side is bounded below independently of $m$. Hence $(\log a_m)$ is bounded below, so $(a_m)$ is bounded away from 0. Therefore

$$L > 0.$$

Thus

$$\lim_{m\to\infty} \frac{|GL_m(\mathbb{F}_q)|}{|\operatorname{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{\infty}(1 - q^{-j})$$

exists and is positive. □

*Proof.* Set

$$a_m = \prod_{j=1}^{m}(1 - q^{-j}).$$

Since each factor satisfies $0 < 1 - q^{-j} < 1$, we have $a_m > 0$ for all $m$. Also,

$$a_{m+1} = a_m(1 - q^{-(m+1)}) < a_m,$$

so $(a_m)$ is decreasing. Hence $(a_m)$ is decreasing and bounded below by 0, therefore it converges. □